GOVERNMENT OF TELANGANA
ELECTRICAL INSPECTORATE

From:
The Deputy Chief Electrical Inspector to
Hyderabad Rural , 5TH Floor, R&B Buildings,
A.C Guards Masabtank,
Hyderabad-500 004

TO:
M/s Genix Polymers Pvt Ltd,
159,TIF Green Industrial Park,
Choutuppal,Yadadri Dist.

Sir,

Lr.No.DYCEIG/TS/EI Ex. 650V/HV/HT-YDD-2024-20056/D.No:461/24 Dated: 26-09-2024

Sir,

Sub: The Electricity Act-2003 and Central Electricity Authority (Measures relating to safety and Electric Supply) Regulations, 2010 – Electrical Installation of Voltage Exceeding 650 V of M/s Genix Polymers Pvt Ltd,159,TIF Green Industrial Park, Choutuppal, Yadadri District. Approval of Drawings, submitted towards Design Approval-Under Section 54 of The Electricity Act-2003 and Regulation – 45(3) of CEA (Measures relating to Safety and Electric Supply) Regulations, 2023– Accorded – Regarding.

Ref: - Management letter received dated.26-09-2024

****

The drawings submitted for design approval by you in your letter cited are provisionally approved and valid up to 26-09-2025 subject to the compliance of conditions specified in the annexure to the drawings. The drawings approved for the erection of Equipment as per standard.

The defects/omission pointed in the drawings will be specifically inspected by the Inspecting Officer/DY. Chief Electrical Inspector to Government at the time of Inspection. The Inspection fee of your exceeding 650V installation works out to Rs7800 /- (Rupees Seven Thousand eight hundred only)

After completing the installation to your satisfaction, you are requested to submit the reports for inspection under regulation 45(3) Central Electricity Authority (Measures relating to Safety and Electric Supply) Regulations, 2023.

The consumer shall provide the evidence that the supplier also completed his works in full shape, at the time of request for inspection of the installation.

The installation shall confirm to the relevant provisions of Central Electricity Authority (Measures relating to Safety and Electric Supply) Regulations, 2023 and Bureau of Indian Standards specifications.

Yours faithfully,

RAMANJANEYULU CHALAMALASETTI

Digitally signed by RAMANJANEYULU CHALAMALASETTI
Date: 2024.09.26 18:02:50 +05'30'

DEPUTY CHIEF ELECTRICAL INSPECTOR
HYDERABAD(RURAL)

ANNEXURE

Annexure to the Lr.No.DyCEIG-Hyd-Rural/TG/EI Ex. 650V//HT-YDD-2024-20056/D.No. 461/2024 Dated: 26-09-2024Pertaining to M/Genix Polymers Pvt Ltd,159,TIF Green Industrial Park, Choutuppal,YadadriDist

I POINT OF COMMENCEMENT OF SUPPLY:

For the H.T. over head line if any, from the point of commencement of supply, guarding shall be provided underneath the lines and the guards shall be efficiently earthed.

II PREFABRICATED SUB-STATION

The service conditions, rated characteristics, general structural requirements and test methods of prefabricated substations shall be as per IS:14786 (or) IEC:62271-202

III TRANSFORMER:

1. The Transformer plinth shall be constructed by the side of the D.P. Structure in the opposite direction of incoming H.V., supply at a distance between 1.5 to 2.0 Meters. (Regulation 12)
   The Transformer rating and capacity shall conform as per IS1180 (Part-IV):2014
2. Inter-locks shall be provided between the isolators and their respective circuit breakers and earth switches.
3. When two or more transformers are operated in parallel, the system shall be so arranged as to trip the secondary breaker of a transformer in case the primary breaker of that transformer trips. [Regulations 45]
4. Earth electrodes of neutral of the Transformers/Generators and lighting arrestors normally shall not be inter-connected.
5. Copper shall be used for Neutral and lighting arrestors earthing.
6. The circuit breaker shall be installed within 15 meters of walking distance.
7. The Bus duct shall be provided on the secondary side of the 750 KVA and above capacity Distribution Transformer.
8. Where the Transformer capacity exceeds 630 KVA, a circuit breaker of adequate capacity shall be installed on the secondary side of the Transformer.
9. **For oil filled transformers of capacity and voltage up to and including 2500 KVA, 33 KV, only IS certified as per IS: 1180 (Part-I):2014 shall be used as per Electrical Transformers (Quality Control) Order,2015.**

IV CONTROL ROOM:

1. The cables shall be dressed properly and fixed with inverted “U” clamps on the racks that are run in the cable trenches.
2. Cables of different voltages shall be run at different levels. Cables of A.C. and D.C. shall be separated.
3. All the hinged doors in the panel shall be earthed to their respective frames with flexible wires.
4. Identification, such as name of controlling equipment, cable size, length and load shall be noted at their respective control switches on the panel board.

V MACHINERY SECTION:

1. The parallel earth bus shall be run all through the machinery sections with raiser at each equipment or machine. All motors, starters, switches and frames of the machines shall be provided with two body earths, each directly tapped from the two parallel earth buses. The earth wires so connected shall be less than 15ft in length and the size shall be as per the prescribed norms.
2. Normally bare copper or G.I. wire/flat shall be used for body earthing.
3. D.C. Machinery earthing system shall be of copper and shall not be mixed up with A.C., earthing system.
4. The long/cross travel and lifting motors of the E.O.T. Cranes shall be provided with earthing all along the rails earthed at both the ends with adequate size and continuous earth wire/flat.
5. Automatic P.F. controls shall be installed for the capacitors. OR
   Capacitors of adequate capacity shall be connected in shunt for each motor of capacity 20 HP and above. For motors of lesser capacity, consolidated capacitor bank shall be connected at the nearest to the distribution panel.

VI GENERATORS:

1. When two are more generators are operated in parallel an automatic arrangement shall be made such that only one of the neutral earth links of the generators is closed. Due to any fault if one generator fails, then neutral contact of another generator shall be closed automatically.
2. If the generators are to be run in parallel with Suppliers’ grid, arrangement shall be made such that none of the neutral earth links of the generators are closed and for any reason, if Suppliers’ grid fails, the said arrangement shall be such that neutral link of one of the generators shall be closed automatically.

VII GENERAL:

1. The location of D.B.,s shall conform to Regulations 35.
2. Installation of transformers shall be as per I.S. 10028
3. Suitable provisions shall be made for immediate and automatic discharge of every Capacitors on disconnection of supply.
4. All plug sockets shall be of three pin type and the third pin shall be permanently and efficiently earthed.
5. Local control isolators shall be provided for the motors, wherever necessary and any instructions issued in this regard at the time of inspection shall be complied with.
6. Earth pits shall be constructed and maintained as per B.I.S.3043-1987.
7. Outdoor substation except pole type sub station shall be efficiently protected by fencing not less than 1.8 Meters in height or other means so as to prevent access to the electric supply lines and apparatus therein by unauthorized persons. {Regulation 49(ii)}
8. The earthing lead from (both ends of) the lightning arrester should not pass through any iron or steel pipe but shall be taken as directly as possible from the lightning arrester to a separate earth electrode subject to the avoidance of bends where ever practicable. { Regulation 74 (2)}
9. The bottom most portion of the insulator of live part should be at a height, not less than 2.75 m from the ground level.(Regulation12).
10. Separate premises with gate to have access to his equipment from outside at all times shall be provided to the supplier. (Regulation 45)
11. Clearances from the building to that of not exceeding 650 volts lines and service lines shall conform to Regulations 60 of CEA (Measures relating to Safety and Electric Supply) Regulations, 2010 .
12. The overhead line shall be protected with a device for rendering the line electrically harmless in case it breaks. (Regulation 73 (1))
13. An efficient means should be adopted to divert to earth any electrical surges due to lightning on every overhead line. (Regulation 74(1))
14. Panel Boards shall be made of dust and vermin proof.
15. The final sub-circuits of lighting shall not exceed 800 W or 10 points.
16. The Electrical installation work shall be executed only through a Licensed Electrical Contractor as per Regulations 29 (1) of CEA (Measures relating to Safety and Electric Supply) Regulations, 2010 .
17. The Electrical contractor shall submit his commencement report to this office immediately after commencing the Electrical Installation work duly indicating the name of the permit holders supervising and executing the work.
18. The Licensed Electrical Contractor who has executed the work shall submit his completion-cum-test report in the prescribed proforma after completing the work. And the person who has executed the work shall be present at the time of inspection.
19. The load on each generator shall be restricted to its capacity.
20. Identifications such as load particulars, item No's on switches starters, motors etc., as shown in the drawings and lists produced shall be marked on the panel boards.(Regulation 14).
21. All the portable and transportable equipment shall be connected to supply with flexible cable enclosed in a metallic covering with continuity. The earth wire of the cable shall be connected to one of the body earths. The Second body earth shall be earthed through the flexible covering.
22. The 4 pole change over switch and two independent mechanical isolators one for DISCOM supply and another for Generator shall be provided on either side of the changeover switch.
23. One meters clearance shall be maintained around the Generator and in-front of switch Boards.
24. No service line or tapping shall be taken off an overhead line except at a point of support (Regulations 71).
25. New register shall be opened and the values of the earth resistance of the various earth electrodes and insulation resistance values of the transformer shall be entered in the registers.
26. The Licensed contractor who executed the work and the authorized person must be present at the time of Inspection.
27. The consumer is advised to get the installation inspected within one year from the date of approval of drawings.
28. Approved drawings shall be displayed in the Factory Premises.
29. This approval is provisional and valid up to one year from the date of approval. Supply shall be obtained only after obtaining the statutory approval under Regulation 45(3) from the Deputy Chief Electrical Inspector , Rural, Hyderabad.

Yours faithfully,

RAMANJANEYULU CHALAMALASETTI

Digitally signed by RAMANJANEYULU CHALAMALASETTI
Date: 2024.09.26 18:03:19 +05'30'

Deputy Chief Electrical Inspector , Rural,
Hyderabad .

Follow the Electricity Act & Regulations of CEA to avoid Electrical Fatal and Fire Accidents

APPROVED

Subject to the condition laid down in this office

Lr No Dy.CEIG/Hyd/R/T S/HT/MSB/D No ———

Dt ——— Office Reg No ———

The Electrical Installation shall be laid as per the approved Drawings. Any deviations shall be got approved earlier.

RAMANJANEYULU CHALAMALASETTI

Digitally signed by RAMANJANEYULU CHALAMALASETTI
Date: 2024.09.26 18:03:38 +05'30'

By Chief Electrical Inspector to Govt
Hyderabad (Rural)

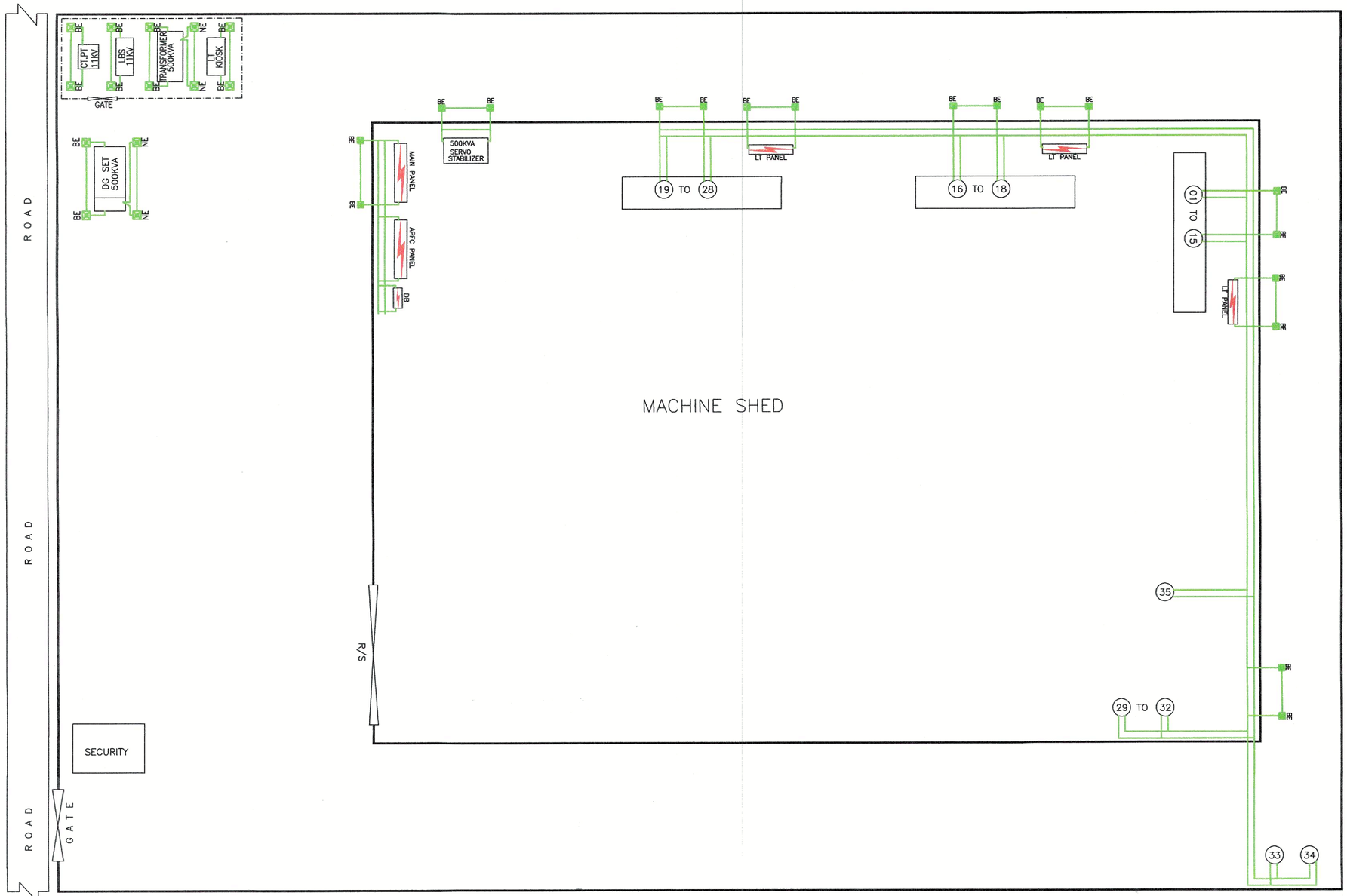
UNITED POWER LINKS
"A" Grade Electrical Contractors
Lic. No. A2-2736
T-2381
9849518088, 9000046469

CLIENT:- M/s.GENIX POLYMERS PVT LTD

PLOT.NO:159, IP DANDUMALKAPUR(V), CHOUTUPPAL(M), YADADRI DISTRICT, TELANGANA.

TITLE:- SINGLE LINE DIAGRAM

DWG.No:001	DATE: 22.09.2024	SCALE: NTS	DRAWN:P.V.RajeshKumar



SITE AND EQUIPMENT LAYOUT

APPROVED

subject to the condition laid down in this office

Lr No.Dy.CEIG/Hyd/R/T S/HT/MSB/D No. _____

Dt _____ Office Reg No _____

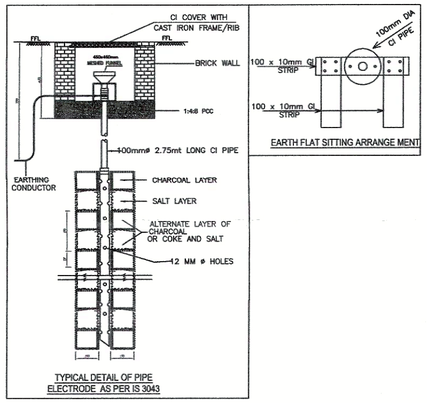
The Electrical Installation shall be laid as per the approved Drawings. Any deviations shall be got approved earlier.

RAMANJANEYULU CHALAMALASETTI

Digitally signed by RAMANJANEYULU CHALAMALASETTI
Date: 2024.09.26 18:03:50 +05'30'

Dy Chief Electrical Inspector to Govt
Hyderabad (Rural)

SL.NO	NAME OF EQUIPMENT	LOAD	
	DOUBLE WALL PIPE CORRUGATORS-A		
1	EXTRUDER DRIVE MOTOR	75	KW
2	HEATING LOAD	12	KW
3	COOLING POWER	2.2	KW
4	MOULD HEATING	14	KW
5	FORWORD BACKWORD MOTOR	1.2	KW
6	OPEN CLOSE MOTOR	0.4	KW
7	VACUUM PUMP	11	KW
8	VACUUM PUMP	11	KW
9	VACUUM PUMP	11	KW
10	OIL PUMP	60	KW
11	FAN MOTOR	7.5	KW
12	SUB MOTOR-1	0.36	KW
13	SUB MOTOR-2	0.4	KW
14	SUB MOTOR-3	2.2	KW
15	SUB MOTOR-4	0.75	KW
	DOUBLE WALL PIPE CORRUGATORS-B		
16	EXTRUDER DRIVE MOTOR	132	KW
17	HEATING LOAD	16	KW
18	COOLING POWER	2.2	KW
	PIPE EXTRUSION LINE		
19	EXTRUDER DRIVE MOTOR	160	KW
20	HEATING LOAD	40	KW
21	COOLING POWER	1	KW
22	FORWORD BACKWORD MOTOR	1.1	KW
23	POWER MOTOR	5.5	KW
24	VACUUM PUMP	5.5	KW
25	WATER PUMP	11	KW
26	WATER PUMP	5.5	KW
27	POWER MOTOR	5.5	KW
28	SERVO MOTOR	0.7	KW
	CHILLER		
29	CHILLER COMPRESSOR	44.5	KW
30	CHILLER COMPRESSOR	44.5	KW
31	FAN MOTOR	1.5	KW
32	PUMP	3.75	KW
	RO PLANT		
33	ID FAN	7.5	KW
34	PUMP	5.5	KW
35	AIR COMPRESSOR	22.5	KW
36	LIGHTING LOAD	5	KW



NOTES:–

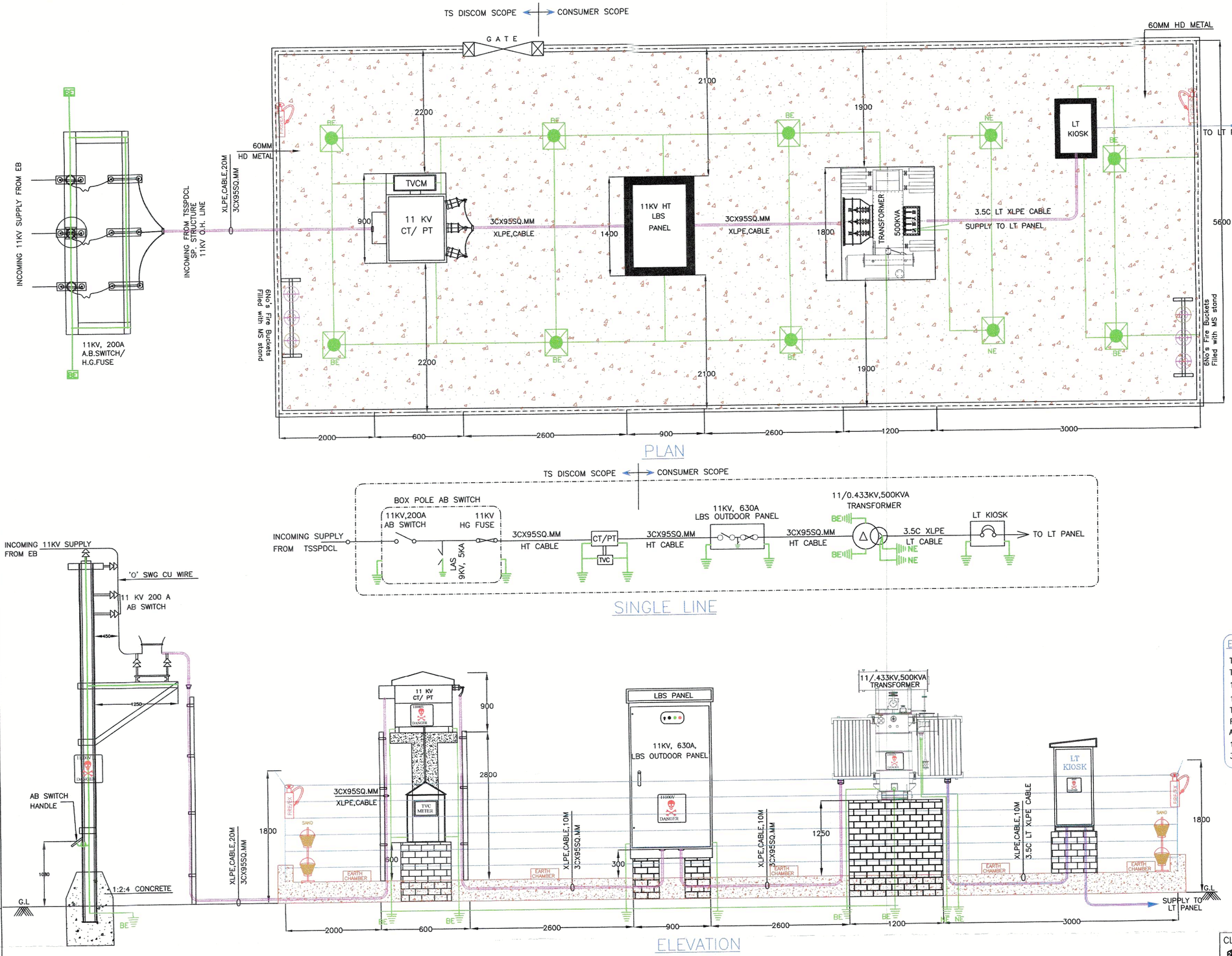
1. ALL EQUIPMENTS SHALL BE DOUBLE EARTHED WITH ADEQUATE SIZE OF Cu.FLATS/WIRE.
   * MAIN EARTH BUS – 50x6MM G.I. FLAT 2R.
   * PANELS – 50X6MM G.I. FLAT 2R.
   * EQUIPMENTS UPTO 15HP – 8SWG G.I. WIRE 2R.
   * EQUIPMENTS ABOVE 15HP – 25X3MM G.I. FLAT 2R.
2. 1525mm CLEARANCE WILL BE MAINTAINED BETWEEN EARTH PIT AND WALL.
3. 3050mm CLEARANCE WILL BE MAINTAINED BETWEEN TWO EARTH PITS.
4. ALL THE EARTH PITS ARE INTER LINKED BY 50x6mm G.I. FLAT.
5. LIGHTNING ARRESTOR EARTHING BY 25x3 mm Cu. FLAT.
6. FENCING FRAME EARTHING IS PROVIDED WITH 8 SWG G.I. WIRE.
7. TRF BODY ARE DOUBLE EARTHED WITH 75x6 mm GI. FLAT.
8. TRF NEUTRALS ARE DOUBLE EARTHED WITH 40x6 mm Cu. FLAT.
9. DG BODY ARE DOUBLE EARTHED WITH 75x6 mm GI. FLAT.
10. DG NEUTRALS ARE DOUBLE EARTHED WITH 40x6 mm Cu. FLAT.
11. FOLLOWING CLEARANCE WILL BE MAINTAINED DURING THE ERECTION OF SB's, DB's, PANEL's.
    * FRONT CLEARANCE WILL BE MORE THAN 1000MM. AT REAR/SIDE.
    * REAR CLEARANCE WILL BE MORE THAN 750MM IF THERE ARE CABLE ENTRIES
    * REAR CLEARANCE WILL BE LESS THAN 200MM IF THERE IS NO CABLE ENTRIES AT REAR/SIDE.
    * FOR WALL MOUNTING BOARDS CABLE ENTRIES ARE TOP AND BOTTOM ENTRY ONLY NOT AT REAR/SIDE.

UNITED POWER LINKS
"A" Grade Electrical Contractors
Lic. No. A2-2736 T-2381
9849518088, 9000046469

CLIENT:–
M/s.GENIX POLYMERS PVT LTD
PLOT.NO:159, IP DANDUMALKAPUR(V), CHOUTUPPAL(M), YADADRI DISTRICT, TELANGANA.

TITLE:– SITE & EQUIPMENT LAYOUT

DWG.No:002	DATE: 22.09.2024	SCALE: NTS	DRAWN: P.V.RajeshKumar



APPROVED

Subject to the condition laid down in this office

Lr No Dy.CEIG/Hyd/R/T S/HT/MSB/D No ——

Dt —— Office Reg No ——

The Electrical Installation shall be laid as per the approved Drawings. Any deviations shall be got approved earlier

RAMANJANEYULU CHALAMALASETTI

Digitally signed by RAMANJANEYULU CHALAMALASETTI
Date: 2024.09.26 18:04:02 +05'30'

By Chief Electrical Inspector to GOVT
Hyderabad (Rural)

EARTHING DETAILS:

TRANSFORMER BODY EARTHING WITH 2Rx75x6mm GI STRIP
TRANSFORMER NEUTRAL EARTHING WITH 2Rx40x6mm Cu. STRIP
11KV LBS BODY EARTHING WITH 2Rx50x6mm GI STRIP
11KV, CT/PT BODY EARTHING WITH 2Rx50x6mm GI STRIP
TVC METER BOX EARTHING WITH 2Rx25x3mm GI STRIP
FENCING EARTHING WITH 2Rx50x6mm GI STRIP
ALL EARTH ELECTRODES ARE 75mm DIA x 2.75Mtrs. LENGH CI PIPE
10' DISTANCE MAINTAINED BETWEEN TWO EARTH ELECTRODES
3' DISTANCE MAINTAINED ANY FOUNDATION TO EARTH ELECTRODES

UNITED POWER LINKS
"A" Grade Electrical Contractors
Lic. No. A2-2736
T-2381
9849518088, 9000046469

CLIENT:–
M/s.GENIX POLYMERS PVT LTD
PLOT.NO:159, IP DANDUMALKAPUR(V), CHOUTUPPAL(M), YADADRI DISTRICT, TELANGANA.

TITLE:–
11KV/433V HT/LT SWITCH YARD

DWG.No:003	DATE: 22.09.2024	SCALE: NTS	DRAWN:P.V.RajeshKumar